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Full Length Article

Hierarchical cross-attention guided deformable registration with multi-level feature fusion for medical images

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ABSTRACT

Deformable medical image registration is fundamental for longitudinal diagnosis, surgical planning, and quantitative anatomical analysis, yet achieving accurate and robust alignment remains challenging. Traditional optimization-based methods are computationally intensive and highly sensitive to parameter settings. In contrast, many recent deep learning approaches fail to effectively capture spatial correspondences across diverse anatomical structures, struggle with multi-scale feature fusion, and lack interpretability. To address these issues, we propose Hierarchical Cross-Attention Morph (HCA-Morph), a novel unsupervised registration framework that explicitly models spatial alignment and adaptively integrates hierarchical features. HCA-Morph comprises two dedicated modules: a Spatial Correspondence-Aware Module (SCAM), which learns interpretable spatial alignment between moving and fixed images from local to global scales, and a Cross-Scale Attention Module (CSAM), which dynamically fuses multi-level features with built-in interpretability to enhance representation consistency. Experiments on two public brain MRI datasets-the Open Access Series of Imaging Studies (OASIS) and Information eXtraction from Images (IXI)-demonstrate that HCA-Morph achieves Dice scores of 0.845 and 0.834 on the OASIS and IXI datasets, respectively, outperforming current state-of-the-art models, while also achieving better HD95 scores (1.365 on OASIS and 1.725 on IXI). Notably, with only 2.05 MB parameters-the smallest among all compared methods-HCA-Morph achieves comparable or superior accuracy while maintaining the lowest memory consumption (4.08 GB on OASIS, 9.02 GB on IXI) and generating more plausible deformation fields with fewer folding regions ($|J_\phi \leq 0|$: 1.77×10^{-4} on OASIS, 5.39×10^{-6} on IXI). Ablation studies confirm that SCAM and CSAM provide complementary contributions on OASIS, combining synergistically to yield a total improvement of 0.030. Portability experiments further demonstrate that both modules can be successfully integrated into diverse existing architectures including CNN, Transformer, and Mamba-based models, delivering consistent performance improvements across different backbones. Overall, HCA-Morph achieves a strong balance between accuracy, efficiency, and anatomical plausibility for deformable brain MRI registration. The source code and pretrained models are publicly available at https://github.com/Importes/HCA-Morph.

1. Introduction

Medical image registration is a cornerstone of modern image-based medicine, aiming to spatially align scans acquired at different times, from different scanners, or across different patients so that corresponding anatomical structures can be accurately matched. This process underpins a broad spectrum of clinical and research applications, including monitoring disease progression in longitudinal studies, aligning intraoperative images with preoperative scans for surgical navigation, and improving treatment planning as well as postoperative follow-up (Deng et al., 2023). Compared with natural images, medical scans exhibit substantial pathological variability and inter-subject anatomical heterogeneity, particularly within complex cortical and subcortical regions (Kuang & Schmah, 2019). Accurate registration therefore demands nonlinear, dense deformation fields that can capture fine-grained anatomical correspondences. Classical algorithms such as SyN (Avants et al., 2008) and Elastix (Klein et al., 2010) address this challenge through iterative optimization of similarity metrics. While effective, these approaches are computationally intensive, sensitive to hyperparameter tuning, and often struggle to generalize across diverse clinical scenarios.

Recently, unsupervised deep learning techniques have emerged as powerful alternatives, enabling end-to-end estimation of deformation fields with markedly improved efficiency and predictive accuracy. Despite their success, current learning-based methods still face two major limitations: insufficient modeling of cross-scale semantic dependencies

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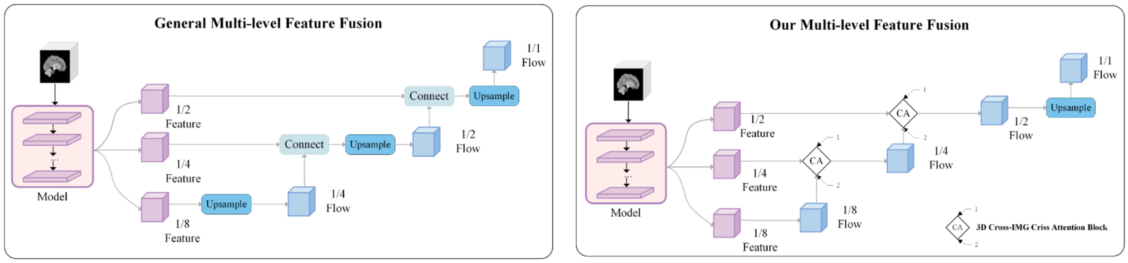


(a) (b)

Fig. 1. Comparison of general multi-level feature fusion (a) and our multi-level feature fusion (b). Traditional pyramid fusion (a) uses static upsampling and concatenation, treating all features equally. Our design (b) allows each level to adaptively query other levels for relevant features via cross-attention (CA), strengthening informative signals while suppressing noise. Superscripts and subscripts denote the direction of attention flow between levels. Detailed mechanisms are discussed in Section 2.2.3.

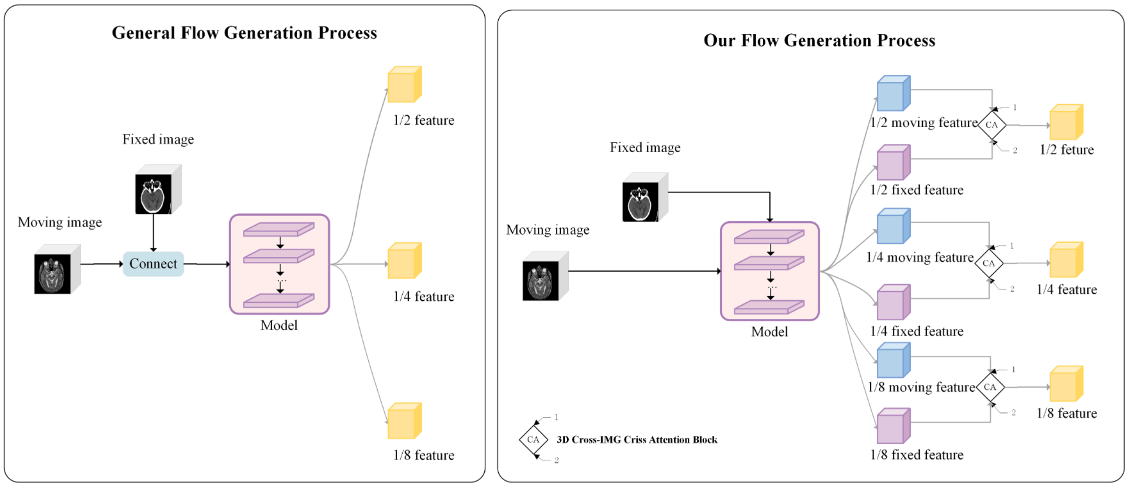
and the absence of explicit correspondences between moving and fixed images.

For multi-level feature fusion, pyramid-based (Kang et al., 2022) and UNet-based (Çiçek et al., 2016) architectures are widely employed. As illustrated in Fig. 1a, cross-level integration is typically realized through upsampling and skip connections. However, these approaches treat all features uniformly-a low-level edge feature and a high-level anatomical feature receive equal weight, even when their relevance varies greatly across regions. This “one-size-fits-all” strategy can cause critical details (e.g., subtle boundary changes in atrophic regions) to be diluted as features propagate from fine to coarse scales (Liu et al., 2019; Wang et al., 2021). To overcome this limitation, as shown in Fig. 1b, we introduce a top-down attention mechanism that adaptively selects which features to emphasize at each level, explicitly highlighting informative representations while suppressing redundant responses-similar to how radiologists focus on specific anatomical regions when comparing scans.

For spatial correspondence fusion, as illustrated in Fig. 2a, simple channel stacking of the moving and fixed images provides only an implicit coupling, requiring the network to infer anatomical correspondences indirectly through local convolutions. In other words, the network must “guess” which voxels in the moving image correspond to which voxels in the fixed image, without explicit guidance. Such an implicit design may lead to distorted or ambiguous spatial relationships between the two images (Sivan et al., 2023; Song et al., 2022). To overcome this limitation, as shown in Fig. 2b, we introduce an attention mechanism that directly computes voxel-to-voxel correspondences, thereby capturing spatial dependencies from local to global scales in an explicit and data-driven manner.

In 2022, Chen et al. proposed TransMorph (Chen et al., 2022), which employs explicit channel-wise fusion of moving and fixed images, a pyramid-style architecture for inter-level feature fusion, and a Transformer for feature extraction. In this approach, the model is required to implicitly learn spatial correspondences between moving and fixed images, while the reliance on a pyramid-like structure may lead to error propagation across levels. In the same year, Shi et al. introduced XMorpher (Shi et al., 2022), which uses early channel concatenation to form a joint representation-a simple design that nevertheless requires convolution layers to implicitly infer spatial correspondences. In 2024, Wang et al. proposed the Pyramid Attention Network (PAN) (Wang et al., 2024a), which performs spatial correspondence fusion by letting the warped moving image attend to the fixed image, followed by a pyramid-like scheme where the deformation field estimated at each stage is progressively upsampled and passed to the next stage for further registration; while effective, this design may amplify errors propagated from inaccurate lower-level deformation fields. In the same year, Li et al. introduced GaussianDIR (Li et al., 2024), which employs sparse Gaussian kernels for moving–fixed fusion, offering interpretability and efficiency but at the cost of higher implementation complexity, while Ma et al. proposed LLaMA-Reg (Ma & Yang, 2024), which leverages LLaMA for cross-image fusion. Also in 2024, Meng et al. presented CorrMLP (Meng et al., 2024), a state-of-the-art framework where moving and fixed images are first processed independently by convolutional encoders and then fused through channel-level concatenation; for multi-level feature fusion, it adopts a convolution-like paradigm in which the deformation field estimated at one stage is repeatedly applied to the moving features of the subsequent stage, though this strategy often produces features contaminated with irrelevant information. Most recently, in 2024-2025, Wang et al. proposed MambaMorph (Wang et al., 2025) and VMambaMorph (Wang et al., 2024b), which leverage state-space dynamics to enhance long-range modeling and computational efficiency, yet still rely on pyramid-like fusion strategies.

Building on prior work, we observe that although many innovative methods have been proposed, spatial correspondence fusion is still often handled by simple channel concatenation or implicit attention mechanisms, overlooking the intrinsic anatomical correspondence between the moving and fixed images. Likewise, multi-level feature fusion is frequently implemented using pyramid-like structures, which are prone to cross-level error propagation. To address these limitations, we present Hierarchical Cross-Attention Morph (HCA-Morph), a deformable registration framework that jointly models progressive multi-level semantics and anatomically grounded cross-image correspondence. The central question driving this study is whether explicit cross-attention and cross-scale attention modules can enhance anatomical correspondence modeling and multi-level feature integration, thereby substantially improving overall registration performance while ensuring stable portability and incremental benefits across diverse baselines. HCA-Morph comprises two synergistic components: (1) a Cross-Scale Attention Module (CSAM) that adaptively enables low- and high-level features to attend to each other, preserving semantic flow across layers; and (2) a Spatial Correspondence-Aware Module (SCAM) that constructs a deformable attention field between the moving and fixed images, replacing rigid concatenation with anatomically aligned mappings. We compare HCA-Morph with five recent models (PAN, TransMorph, CorrMLP, MambaMorph, and VMambaMorph) on two public Magnetic Resonance Imaging (MRI) brain datasets, OASIS (LaMontagne et al., 2019) and IXI (London, 2023), and summarize their key architectural distinctions in Table 1. Table 1 reveals a critical architectural distinction: HCA-Morph is the only framework that employs explicit cross-attention for both spatial correspondence and cross-scale fusion, whereas all baseline methods rely on at least one implicit mechanism (requiring the network to infer correspondences indirectly) or pyramid-based fusion (which may propagate errors across levels). This dual-explicit design translates to measurable advantages: SCAM contributes to performance improvements on both OASIS and IXI by directly modeling interpretable voxel-to-voxel correspondences, while CSAM enhances registration through adaptive cross-scale attention with inherent interpretability that prevents detail loss. Critically, their joint effect (+0.030 on OASIS, +0.010 on IXI) substantially exceeds their individual contributions,



(a) (b)

Fig. 2. Comparison of flow generation processes. (a) General pipeline with channel-wise concatenation, where correspondences are inferred implicitly. (b) Our design with deformable 3D Cross-IMG-Criss Attention Block (CA) between moving and fixed features, explicitly preserving anatomical alignment across scales. Superscripts denote attention from finer to coarser levels.Detailed 3D Cross-IMG-Criss Attention Block mechanisms are discussed in Section 2.2.3.

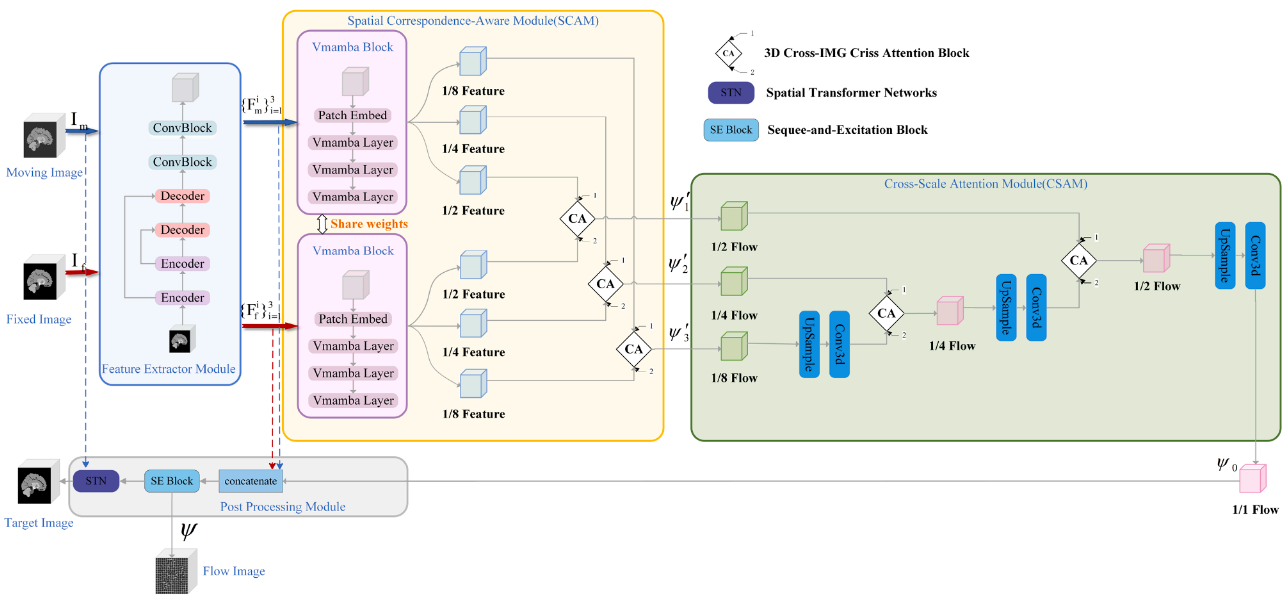


Fig. 3. Overall framework of HCA-Morph. The model consists of four modules: Feature Extractor, Spatial Correspondence-Aware Module (SCAM), Cross-Scale Attention Module (CSAM), and Post-Processing. SCAM explicitly aligns moving and fixed features, while CSAM fuses multi-level deformation fields. Numbers 1 and 2 indicate cross-attention direction across levels (from level 1 to level 2).

confirming that explicit modeling of both spatial and hierarchical correspondences is essential. The inherent interpretability of both modules-SCAM through traceable spatial alignment patterns and CSAM through visualizable attention weights-enables transparent inspection of registration decisions. Overall, HCA-Morph achieves Dice scores of 0.845 (OASIS) and 0.834 (IXI), outperforming the strongest baseline (VMambaMorph), while maintaining the lowest memory footprint (4.08 GB and 9.02 GB).

Our contributions are summarized as follows:

- We propose HCA-Morph, an interpretable deformable registration framework achieving Dice scores of 0.845 (OASIS) and 0.834 (IXI), with HD95 values of 1.365 and 1.725, respectively. The method requires only 4.08 GB memory-the lowest among all compared approaches-and the improvements are statistically significant.
- We introduce the Cross-Scale Attention Module (CSAM) for adaptive multi-level feature fusion via visualizable attention, and the Spatial Correspondence-Aware Module (SCAM) for explicit voxel-wise alignment through local-to-global cross-attention.
- Joint integration of CSAM and SCAM yields improvements of 0.030 (OASIS) and 0.010 (IXI), exceeding their individual effects. Both modules generalize across CNN-, Transformer-, and Mamba-based architectures.

2. Methodology

Medical image registration seeks a deformation field ψ that warps the moving image I_m onto the fixed image I_f, establishing voxel-wise anatomical correspondence. We parameterise ψ using a deep neural network that outputs a dense deformation grid for resampling I_m. As shown in Fig. 3, HCA-Morph consists of four components: a Feature Extractor Module, a Spatial Correspondence-Aware Module (SCAM), a Cross-Scale Attention Module (CSAM), and a Post-Processing Module. Both input images share encoder weights to promote joint feature extraction.

Table 1
Architectural feature matrix for the compared methods. **Legend:** ✓ = explicitly present; ∘ = implicit/partial implementation; × = absent. **Spatial correspondence fusion:** *cross-attn* = cross-attention directly between moving and fixed images; *channel* = channel-wise concatenation. **Multi-level feature fusion:** *cross-scale attn* = attention across different resolutions; *FPN-like* = pyramid-style top-down fusion with lateral connections. Note: HCA-Morph is the only method employing explicit cross-attention (✓) for both spatial and cross-scale fusion, avoiding implicit coupling and pyramid-based error propagation. Detailed mechanisms are discussed in Section 3.4.1.

Method	Backbone	spatial correspondence fusion		multi-level feature fusion	
		cross-attn	*channel*	*cross-scale attn*	*FPN-like*
HCA-Morph (ours)	Vision Mamba (VMamba)	✓	×	✓	×
PAN (Wang et al., 2024a)	CNN	∘	×	×	✓
TransMorph (Chen et al., 2022)	Transformer	×	✓	×	✓
CorrMLP (Meng et al., 2024)	CNN + MLP	×	∘	×	✓
MambaMorph (Wang et al., 2025)	Mamba	×	∘	×	✓
VMambaMorph (Wang et al., 2024b)	VMamba	×	∘	×	✓

Table 2
Architectural hyperparameters of HCA-Morph.

Module	Item	Amount	Notes
Feature Extractor	Encoder	2	Hierarchical feature encoding
	Decoder	2	Hierarchical feature decoding
	ConvBlock	2	Refinement of extracted features
SCAM	VMamba Block	1	Volumetric spatial-state modeling
	VMamba Layer	3 pre VMamba Block	Multi-resolution representation learning
	3D VSS Block	4 per VMamba Layer	Multi-resolution representation learning
	3D Cross-IMG-Criss Attention Block	1	Explicit spatial correspondence fusion
CSAM	3D Cross-IMG-Criss Attention Block	1	Multi-level feature fusion
	Conv3d	2	Global feature refinement
	Upsample	2	Feature Upsample
Post Processing	SE Block	1	fuse ψ_0 with the feature maps of the moving and fixed images(F_m *and* F_f)
	STN	1	Apply the deformation field to the moving image to complete the registration

After feature extraction, F_m and F_f are downsampled via patch embedding and processed by three VMamba layers (Liu et al., 2024), yielding multi-level features F_m^1, F_m^2, F_m^3 and F_f^1, F_f^2, F_f^3. A 3D Cross-IMG-Criss Attention Block refines fixed-image features using moving-image features at each scale ($F_m^3 \rightarrow F_f^3$, $F_m^2 \rightarrow F_f^2$, $F_m^1 \rightarrow F_f^1$), generating preliminary deformation fields ψ_3', ψ_2', and ψ_1' in parallel. CSAM then fuses these multi-level fields: ψ_2' attends to ψ_3' to yield ψ_2'', then ψ_1' attends to ψ_2'' to produce the final field ψ_0. Low-level fields provide global context while high-level fields preserve fine details.

The Post-Processing Module refines ψ_0 by first concatenating it with original features F_m and F_f, then applying a convolutional block and an SE block (Hu et al., 2018) to obtain smoothed field ψ. An STN (Jaderberg et al., 2015) warps I_m with ψ to produce the registered image. Following TransMorph (Chen et al., 2022), we employ three loss terms: L_{ncc} for image similarity, L_{dsc} for organ overlap, and L_{reg} for deformation smoothness. Details are in Section 2.5. Sections 2.1–2.4 describe the modules, and Table 2 lists hyperparameters.

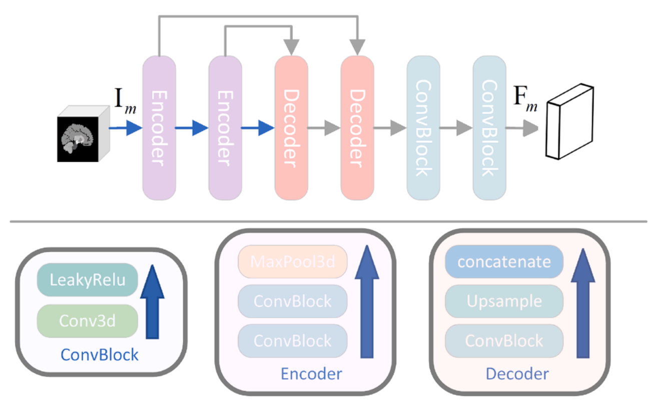


Fig. 4. Simplified 3D U-Net architecture.

2.1. Feature extractor module

We adopt a streamlined 3D U-Net to lift the single-channel inputs into a richer feature space, enabling subsequent modules to extract more informative representations, as shown in Fig. 4. This Feature Extractor Module is used for initial feature extraction. The backbone comprises two encoder and two decoder stages with symmetric skip connections, together with two residual ConvBlocks for feature regression. This minimalist topology balances efficiency with accuracy while preserving fine details. GELU activations are used throughout to stabilize optimization and mitigate vanishing/exploding gradients. Finally, two additional residual ConvBlocks are applied to refine the resulting representations.

More concretely, For an input moving image $I_m \in \mathbb{R}^{b\times c\times h\times w\times l}$, the encoder progressively compress spatial resolution while enriching semantic content; the decoders, equipped with residual skip connections, restore resolution and fuse low- and high-level cues. Two subsequent ConvBlocks integrate these representations to produce the initial feature map $F_m \in \mathbb{R}^{b\times d\times h\times w\times l}$, where $d = 8c$. Applying the same pipeline to the fixed image I_f yields $F_f \in \mathbb{R}^{b\times d\times h\times w\times l}$. Here, b, c, h, w, and l denote the batch size, input channels, height, width, and depth of the 3D image.

2.2. Spatial correspondence-aware module (SCAM)

In spatial correspondence fusion, considering the intrinsic relationships between moving and fixed images, we directly construct a deformable attention field across the two inputs, replacing rigid concatenation with anatomically aligned mappings. Specifically, given feature maps $F \in \mathbb{R}^{b\times d\times h\times w\times l}$ from the Feature Extractor Module (F denotes either F_m or F_f), the SCAM module employs a VMamba Block to generate multi-resolution representations $\{F_m^i\}_{i=1}^3$ and $\{F_f^i\}_{i=1}^3$ for the moving and fixed images, respectively. These hierarchical features are then

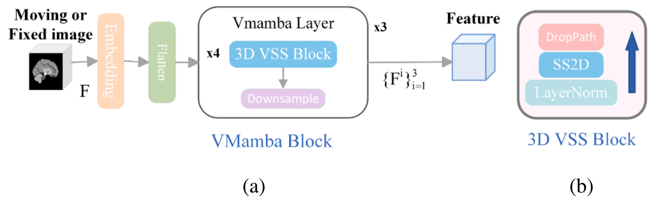


Fig. 5. Structure of the VMamba block and 3D VSS block.

fed into the 3D Cross-IMG-Criss Attention Block to perform spatial correspondence modeling, ultimately yielding the deformable attention-enhanced outputs $\{\psi_i'\}_{i=1}^3$. The architectural hyperparameters and algorithmic workflow of this module are detailed in Table 2 and Algorithm 1.

2.2.1. VMamba block

VMamba (Liu et al., 2024) was originally developed for 2D images, combining the global self-attention of Vision Transformers (ViT) (Dosovitskiy et al., 2021) with the efficiency of convolutional networks. We extend this to 3D images, forming a VMamba Block with three components: a Patch Embedding layer, a Flatten layer, and three VMamba Layers. The Patch Embedding layer reduces feature dimensionality, while the Flatten layer aligns features for VMamba processing. The three VMamba Layers generate feature maps with downsampling rates of 2×, 4×, and 8×. As shown in Algorithm 1 and Fig. 5a, this block reshapes input $F \in \mathbb{R}^{b\times d\times h\times w\times l}$ into output $F \in \mathbb{R}^{b\times(2^i d)\times\frac{h}{2^i}\times\frac{w}{2^i}\times\frac{l}{2^i}}$.

2.2.2. VMamba layer

As shown in Fig. 5a, each VMamba layer consists of four 3D VSS Blocks followed by a PatchMerge operation (except the first layer). Each 3D VSS Block contains four SS2D units (Liu et al., 2024) (Fig. 5b). The 3D VSS Block captures global context without changing spatial resolution, and cascading multiple blocks enables long-range dependency modeling (Wang et al., 2024b). The Embedding and Flatten layer reshapes F into $F \in \mathbb{R}^{b\times k\times 2^i}$, where $k = \frac{h}{2^i}\times\frac{w}{2^i}\times\frac{l}{2^i}$. Four stacked 3D VSS Blocks generate $F \in \mathbb{R}^{b\times 2^i\times\frac{h}{2^i}\times\frac{w}{2^i}\times\frac{l}{2^i}}$. PatchMerge then downsamples each spatial dimension by two, yielding outputs $F_m^i \in \mathbb{R}^{b\times(2^i d)\times\frac{h}{2^i}\times\frac{w}{2^i}\times\frac{l}{2^i}}$ and $F_f^i \in \mathbb{R}^{b\times(2^i d)\times\frac{h}{2^i}\times\frac{w}{2^i}\times\frac{l}{2^i}}$, where i denotes the downsampling level. These hierarchical features are fed into SCAM for spatial correspondence learning.

2.2.3. 3D cross-IMG-criss attention block

To enable the moving image to attend to the fixed image and estimate the deformation field, we extend CrissCrossAttention (Huang et al., 2019) from 2D to 3D, creating the 3D Cross-IMG-Criss Attention Block (Fig. 6). Standard attention mechanisms are computationally expensive and do not naturally support 3D cross-image operations. We replace Conv2d layers with Conv3d, treating moving features as queries (Q) and fixed features as keys (K) and values (V). Queries extract affinity maps from keys, which are aggregated with values and fused with the moving image to produce the attention feature map. Feature maps $F_m^i i = 1^3$ attend to $F f^i i = 1^3$, yielding three scale-specific deformation fields $\psi i' i = 1^3$, where $\psi i' \in \mathbb{R}^{b\times(2^i d)\times\frac{h}{2^i}\times\frac{w}{2^i}\times\frac{l}{2^i}}$.

2.3. Cross-scale attention module (CSAM)

CSAM integrates deformation fields from SCAM across different resolutions. Low-level fields encode broad global context while high-level fields preserve fine anatomical detail. Naïve concatenation often introduces redundancy and boundary blurring. To avoid this, we direct cross-scale attention from high to low level (Fig. 7), allowing detailed fields to selectively absorb global cues without overwhelming local structure.

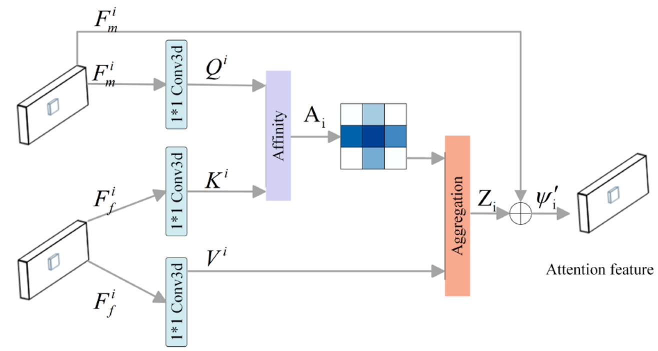


Fig. 6. 3D Cross-IMG-Criss Attention Block, which enables voxel-wise correspondence between moving and fixed images by computing axis-wise affinities along H/W/D and aggregating them into a deformation field.

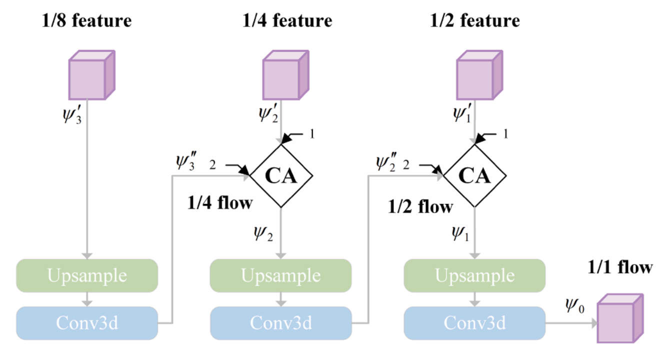


Fig. 7. Architecture of the Cross-Scale Attention Module (CSAM). High-level flows attend to lower-level ones to integrate global context while preserving fine anatomical details.

Two successive CSAM stages fuse the preliminary flows $\psi' i i = 1^3$. After each attention step, features are upsampled and integrated with 3D convolution to obtain the deformation field, preserving both local details and global context for a compact, well-regularized output.

The data flow is summarized in Algorithm 2. SCAM generates three preliminary deformation fields: $\psi' i i = 1^3$, where $\psi' 1 \in \mathbb{R}^{b\times 2d\times\frac{h}{2}\times\frac{w}{2}\times\frac{l}{2}}$, $\psi' 2 \in \mathbb{R}^{b\times 4d\times\frac{h}{4}\times\frac{w}{4}\times\frac{l}{4}}$, and $\psi' 3 \in \mathbb{R}^{b\times 8d\times\frac{h}{8}\times\frac{w}{8}\times\frac{l}{8}}$. First, $\psi' 3$ undergoes upsampling and 3D convolution to yield $\psi'' 3 \in \mathbb{R}^{b\times 4d\times\frac{h}{4}\times\frac{w}{4}\times\frac{l}{4}}$. Then $\psi' 2$ attends to $\psi'' 3$, producing $\psi 2 \in \mathbb{R}^{b\times 4d\times\frac{h}{4}\times\frac{w}{4}\times\frac{l}{4}}$. Upsampling and convolution of ψ_2 generates $\psi'' 2 \in \mathbb{R}^{b\times 2d\times\frac{h}{2}\times\frac{w}{2}\times\frac{l}{2}}$, whereupon $\psi' 1$ attends to $\psi'' 2$ followed by upsampling and 3D convolution to obtain $\psi 0 \in \mathbb{R}^{b\times 2d\times h\times w\times l}$.

2.4. Post processing module

Since the deformation field produced by the HCA (SCAM + CSAM) block, ψ_0, may lack fine high-resolution details, we further refine it by fusing ψ_0 with the feature maps generated by the Feature Extractor Module from the moving and fixed images (F_m and F_f) using a Squeeze-and-Excitation (SE) block. Since these features differ in reliability, the SE block adaptively assigns higher weights to informative channels instead of uniform mixing. To represent displacements along all spatial axes, we replace the SE block's final fully connected layer with a 1×1×1 convolution that outputs three channels, yielding the final displacement field $\psi \in \mathbb{R}^{b\times 3\times h\times w\times l}$. A spatial transformer network (STN) then warps I_m with ψ to produce the registered image.

Algorithm 1 Spatial correspondence-aware module (SCAM).

Input: $F_m, F_f \in \mathbb{R}^{b\times d\times h\times w\times l}$ (features from I_m, I_f)
In this subsection, we denote either F_m or F_f simply by F when describing SCAM. Unfold_α flattens V along axis $\alpha \in \{H, W, D\}$ into per-line sequences; BMM is batched matrix multiplication; Fold_α restores the 3D volume layout.
Output: multi-level flows $\{\psi_i'\}_{i=1}^3$
Notes: γ is a learnable scalar gate.

function VMambaBlock(F)
  for $\ell = 1$ to 3 **do** ▷ 3 VMambaLayers
    $F \leftarrow \mathrm{Embedding}(F)$
    $F \leftarrow \mathrm{Flatten}(F)$
    for $t = 1$ to 4 **do** ▷ each layer: 4 VSS 3D Blocks
      $F \leftarrow$ 3D VSS Block(F)
    end for
    $F \leftarrow \mathrm{Downsample}_{\times 2}(F)$
  end for
  return F
end function
function CCApply(A, V)
  $(A^H, A^W, A^D) \leftarrow \mathrm{Split}(A;\ \mathrm{attn} = [H, W, D])$
  $Z^H \leftarrow \mathrm{Fold}_H\big(\mathrm{BMM}(\mathrm{Unfold}_H(V), (A^H)^\top)\big)$
  $Z^W \leftarrow \mathrm{Fold}_W\big(\mathrm{BMM}(\mathrm{Unfold}_W(V), (A^W)^\top)\big)$
  $Z^D \leftarrow \mathrm{Fold}_D\big(\mathrm{BMM}(\mathrm{Unfold}_D(V), (A^D)^\top)\big)$
  return (Z^H, Z^W, Z^D)
end function
function 3D Cross-Img-Criss Attention Block(F_m^i, F_f^i)
  $Q_i \leftarrow \mathrm{Conv}_{1\times1\times1}(F_m^i)$; $K_i \leftarrow \mathrm{Conv}_{1\times1\times1}(F_f^i)$; $V_i \leftarrow \mathrm{Conv}_{1\times1\times1}(F_f^i)$
  $S_i^H \leftarrow \mathrm{Score1D}_H(Q_i, K_i)$; $S_i^W \leftarrow \mathrm{Score1D}_W(Q_i, K_i)$; $S_i^D \leftarrow \mathrm{Score1D}_D(Q_i, K_i)$
  $A_i \leftarrow \mathrm{Softmax}\big(\mathrm{Concat}(S_i^H, S_i^W, S_i^D)\big)$
  $(Z_i^H, Z_i^W, Z_i^D) \leftarrow \mathrm{CCApply}(A_i, V_i)$; $Z_i \leftarrow Z_i^H + Z_i^W + Z_i^D$
  $\psi_i' \leftarrow \gamma \cdot Z_i + F_m^i$
  return ψ_i'
end function
$\{F_m^i\}_{i=1}^3 \leftarrow$ VMambaBlock(F_m)
$\{F_f^i\}_{i=1}^3 \leftarrow$ VMambaBlock(F_f)
for $i = 1$ to 3 **do**
  $\psi_i' \leftarrow$ 3D Cross-Img-Criss Attention Block(F_m^i, F_f^i)
end for
return $\{\psi_i'\}_{i=1}^3$

2.5. Loss function

The network is optimized with a composite objective that combines the normalized cross-correlation loss L_{ncc} (Zhang et al., 2018), the Dice similarity loss L_{dsc} (Milletari et al., 2016), and a deformation-smoothness regularizer L_{reg} (Balakrishnan et al., 2019). The overall training loss is

$$L_{\mathrm{train}} = \alpha\, L_{\mathrm{ncc}}\big(\tilde{I}_f, I_f\big) \;+\; \beta\, L_{\mathrm{dsc}}\big(\tilde{I}_f^{\mathrm{seg}}, I_f^{\mathrm{seg}}\big) \;+\; \gamma\, L_{\mathrm{reg}}(\psi), \tag{1}$$

where the coefficients α, β, and γ weight the three terms. Here, I_f is the fixed image, $\tilde{I}_f$ is the registered image, I_f^{seg} and $\tilde{I}_f^{\mathrm{seg}}$ are the fixed and registered segmentation masks, and ψ denotes the estimated deformation field.

To compute $\tilde{I}_f^{\mathrm{seg}}$, the moving segmentation I_m^{seg} is first one-hot encoded into K class channels $\{I_{m,k}^{\mathrm{seg}}\}_{k=1}^K$. Each channel is then independently registered by the predicted flow using a spatial transformer network (STN), and the results are concatenated along the channel dimension:

$$\tilde{I}_f^{\mathrm{seg}} = \mathrm{cat}_{k=1}^K\big(\mathrm{STN}(I_{m,k}^{\mathrm{seg}}, \psi)\big), \tag{2}$$

Algorithm 2 Cross-scale attention module (CSAM).

Input: preliminary flows from SCAM: $\psi_1' \in \mathbb{R}^{b\times 2d\times \frac{h}{2}\times\frac{w}{2}\times\frac{l}{2}}$, $\psi_2' \in \mathbb{R}^{b\times 4d\times \frac{h}{4}\times\frac{w}{4}\times\frac{l}{4}}$, $\psi_3' \in \mathbb{R}^{b\times 8d\times \frac{h}{8}\times\frac{w}{8}\times\frac{l}{8}}$
Output: fused flows $\psi_0 \in \mathbb{R}^{b\times d\times h\times w\times l}$
Ops: `Upsample×2` (trilinear, `align_corners=false`);
Stage A: level-2 refinement (high → low).
$\psi_3'' \leftarrow$ `Conv3d`(`Upsample`(ψ_3'))
$Q_2 \leftarrow \mathrm{Conv}_{1\times1\times1}(\psi_2')$; $K_2 \leftarrow \mathrm{Conv}_{1\times1\times1}(\psi_3'')$; $V_2 \leftarrow \mathrm{Conv}_{1\times1\times1}(\psi_3'')$
$\psi_2 \leftarrow$ `3D CROSS-IMG-CRISS ATTENTION BLOCK`(Q_2, K_2, V_2)
Stage B: level-1 refinement (high → low).
$\psi_2'' \leftarrow$ `Conv3d`(`Upsample`(ψ_2))
$Q_1 \leftarrow \mathrm{Conv}_{1\times1\times1}(\psi_1')$; $K_1 \leftarrow \mathrm{Conv}_{1\times1\times1}(\psi_2'')$; $V_1 \leftarrow \mathrm{Conv}_{1\times1\times1}(\psi_2'')$
$\psi_1 \leftarrow$ `3D CROSS-IMG-CRISS ATTENTION BLOCK`(Q_1, K_1, V_1)
$\psi_0 \leftarrow$ `Conv3d`(`Upsample`(ψ_1))
return ψ_0

where $\mathrm{STN}(\cdot)$ applies the displacement field ψ and $\mathrm{cat}(\cdot)$ denotes channel-wise concatenation across classes.

3. Experiments & results

3.1. Datasets and preprocessing

We evaluate HCA-Morph on two widely used public brain MRI registration benchmarks: OASIS (LaMontagne et al., 2019) (link) and IXI (London, 2023) (link). For OASIS, each scan was resampled to 128x128x128 with nearest-neighbor interpolation, IXI was used in its native resolution. no additional preprocessing was performed; the image pairs were used exactly as released, without further modification or augmentation.

OASIS comprises 416 T1-weighted volumes ($160 \times 192 \times 224$ voxels). Two distinct images are then paired, yielding 216 image pairs; 197 are used for training and 19 for testing. Registration quality is assessed on 35 labelled anatomical structures provided with the dataset.

IXI targets atlas-to-patient registration and contains 576 T1-weighted scans at $160 \times 192 \times 224$ resolution. Using the public atlas supplied in TransMorph, we pair the atlas with each subject scan, obtaining 576 pairs. The data are split into 403 training, 58 validation, and 115 test pairs. Performance is measured on 30 anatomical labels.

3.2. Evaluation metrics

To substantiate the effectiveness and superiority of HCA-Morph, we benchmark the model with six complementary metrics: Dice Similarity Coefficient (DSC), 95% Hausdorff Distance (HD95), the Jacobian determinant, i.e., the proportion of voxels with $J_\phi \leq 0$, indicating non-positive local volume changes, inference time (Time), peak GPU memory consumption during inference (Memory), and parameter count (Param).

DSC quantifies the volumetric overlap between the anatomical segmentations of the fixed and the warped moving images within each region of interest; higher values indicate better alignment.

HD95 measures the 95th-percentile surface-to-surface distance between corresponding structures; lower values signify more accurate boundary correspondence.

Jacobian determinant metric reflects the regularity of the predicted deformation field by reporting the proportion of voxels whose local volume change is non-positive (i.e., foldings); a smaller proportion implies a smoother, physically plausible deformation.

Time, **Memory**, and **Param** characterise computational efficiency: shorter runtime, lower memory footprint, and fewer parameters respectively correspond to a model that is faster, lighter, and easier to deploy.

3.3. Runtime environment and training

Our model was developed using PyTorch and was trained for 200 epochs using the Adam optimizer on a machine equipped with an NVIDIA RTX 3090 24GB GPU, an Intel i7-14900KF processor, and 96GB of RAM. The initial learning rate was set to 0.0001 and was updated according to Eq. (3) as the number of training epochs increased. Both the training and inference batch sizes were set to 1.

For the composite loss function, the coefficients α, β, and γ correspond to the NCC, Dice, and deformation regularization terms, respectively. While previous studies often assign equal weights to these components, we empirically observed that the Dice term produced relatively stronger gradient signals (L2 norm $\approx$ 2.2-3.3) compared with NCC ($\approx$ 1.7-2.1) and regularization ($\approx$ 1.4-1.6). As the Dice loss directly reflects anatomical overlap, assigning it a slightly higher weight ($\beta = 2$) helps to balance the optimization gradients, accelerate convergence, and stabilize the training process. Consequently, the coefficients α, β, and γ are set to 1, 2, and 1, respectively.

$$\mathrm{round}\left(0.0001 \times \left(1 - \frac{\mathrm{epoch}_{\mathrm{cur}}}{200}\right)^{0.9}, 8\right) \tag{3}$$

Where $\mathrm{round}(f, 8)$ denotes a floating-point precision of 8 in the f-region and $\mathrm{epoch}_{\mathrm{cur}}$ represents the current training epoch.

3.4. Experimental setup

3.4.1. Comparison with baseline methods

To place HCA-Morph in the context of the current literature, we benchmark it against five representative learning-based registration networks:

CorrMLP: A state-of-the-art registration framework. For spatial correspondence fusion, it adopts an implicit moving–fixed fusion strategy, in which the moving and fixed images are first processed independently by convolutional encoders, and the resulting features are subsequently merged through channel concatenation. For multi-level feature fusion, it employs a pyramid-based design, where the deformation field estimated at the previous stage is iteratively applied to the moving features at the current level, thereby enabling progressive inter-level fusion.

VMambaMorph and **MambaMorph**: Two efficient variants from the Mamba family. For spatial correspondence fusion, both adopt an implicit strategy, in which features extracted from the moving and fixed images are fused via channel stacking. For multi-level feature fusion, both utilize a pyramid-like decoder that progressively upsamples and integrates features through lateral skip connections, ultimately producing the final deformation field.

TransMorph: A Transformer-driven registration baseline. For spatial correspondence fusion, it concatenates the moving and fixed images along the channel dimension and feeds the combined tensor into a Transformer encoder for joint feature extraction. For multi-level feature fusion, it follows a pyramid-based scheme that progressively upsamples and merges features via skip connections across successive levels.

PAN: A recently proposed Pyramid Attention Network. For spatial correspondence fusion, it adopts an implicit moving–fixed fusion strategy, in which the warped moving image attends to the fixed image. For multi-level feature fusion, it employs a pyramid-based architecture, where the deformation field estimated at each stage is progressively upsampled and passed to the next level for refinement, thereby facilitating hierarchical inter-level fusion.

All models are trained from scratch using identical training, validation, and test splits. The baseline configurations follow the official hyperparameter settings provided in their respective papers, ensuring experimental rigor and fair comparison.

Table 3
Ablation settings for evaluating the effects of SCAM and CSAM.

Setting	Description
SCAM-only	Multi-resolution flows generated by SCAM with explicit correspondence, directly fused by a $1 \times 1 \times 1$ convolution (CSAM removed).
CSAM-only	Multi-level fusion performed by CSAM on channel-stacked features, without explicit correspondence modeling (SCAM removed).
None	No SCAM & No CSAM, Features naïvely concatenated; a 3D convolution predicts the flow at each scale followed by upsampling and concatenation-minimal baseline.
HCA (SCAM + CSAM)	The complete HCA-Morph architecture.

3.4.2. Ablation study

To evaluate the effectiveness of CSAM and SCAM, and to examine whether a synergistic interaction exists between them, we conducted a series of ablation experiments under four configurations, isolating the individual and combined contributions of the Spatial Correspondence-Aware Module (SCAM) and the Cross-Scale Attention Module (CSAM) (see Table 3).

SCAM-only. Retains only the Spatial Correspondence-Aware Module to generate multi-resolution deformation fields, which are fused through a simple 1×1×1 convolution without cross-scale refinement.

CSAM-only. Removes SCAM; features from the moving and fixed images are stacked and passed into CSAM, which performs cross-scale feature fusion to regress the deformation field.

NONE. Both SCAM and CSAM are removed. The moving and fixed features are naïvely concatenated, and deformation fields are predicted using 3D convolutions followed by standard upsampling, serving as a minimal baseline.

HCA (SCAM + CSAM). The complete model integrating explicit spatial correspondence modeling (SCAM) with adaptive cross-scale feature fusion (CSAM).

3.4.3. Module-integration experiments

To assess the portability of our modules, we integrate either CSAM or the full HCA block into each baseline and retrain the corresponding models. The resulting designs for spatial correspondence fusion and multi-level feature fusion across different baselines are summarized in Table 4.

CorrMLP with HCA. CorrMLP performs multi-level MLP-based fusion without early concatenation of the moving and fixed images. To preserve this architectural design, we retain CorrMLP's original encoder and append the full HCA block (i.e., both CSAM and SCAM) after feature extraction.

PAN with HCA. Similar to CorrMLP, PAN is extended with the complete HCA block. Its encoder remains unchanged, while SCAM provides explicit spatial alignment between the moving and fixed images, and CSAM replaces the native fusion mechanism to achieve adaptive multi-level integration.

CSAM applied to VMambaMorph, MambaMorph, and TransMorph. These architectures already concatenate the moving and fixed images at the input stage, rendering SCAM unnecessary. Accordingly, we retain their encoders and substitute their native multi-level fusion heads with CSAM, using the CSAM outputs to regress the final deformation field.

3.5. Result and analysis

3.5.1. Comparison with baseline methods

To substantiate the effectiveness and performance advantages of HCA-Morph, we conducted a systematic comparison against five mainstream baselines on the public OASIS and IXI datasets. Evaluation was performed from four complementary perspectives: registration accuracy (Dice coefficient and HD95), physical plausibility of the deformation

Table 4
Design choices for spatial correspondence and multi-level fusion after integrating CSAM or HCA into baseline models. Symbols: ✓present; ∘partial/implicit; ×absent.

Dataset	Method	Module	Spatial correspondence fusion		Multi-level feature fusion	
			cross-attn	*channel*	*cross-scale attn*	*FPN-like*
OASIS	CorrMLP	HCA	✓	×	✓	×
	VMambaMorph	CSAM	×	∘	✓	×
	MambaMorph	CSAM	×	∘	✓	×
	TransMorph	CSAM	×	✓	✓	×
	PAN	HCA	✓	×	✓	×

Table 5
Comparison of registration performance on the OASIS and IXI datasets. Best results are in **bold**, second best are underlined.

Dataset	Method (Year)	Dice ↑	HD95 ↓	$\|J_\phi \leq 0\|$ ↓	Time (s) ↓	Memory (GB) ↓	Param (MB) ↓
OASIS	CorrMLP (2024)	0.831 ± 0.013	1.511 ± 0.210	$1.13 \times 10^{-3} \pm 7.26 \times 10^{-4}$	0.69 ± 0.04	9.60	13.36
	VMambaMorph (2024)	$\underline{0.837 \pm 0.015}$	1.433 ± 0.211	$\underline{1.43 \times 10^{-4} \pm 5.10 \times 10^{-5}}$	$\underline{0.23 \pm 0.04}$	4.22	9.64
	MambaMorph (2025)	0.826 ± 0.017	1.521 ± 0.237	$\mathbf{7.73 \times 10^{-5} \pm 3.16 \times 10^{-5}}$	0.52 ± 0.07	11.60	7.59
	TransMorph (2022)	0.816 ± 0.013	$\mathbf{1.337 \pm 0.221}$	$4.79 \times 10^{-3} \pm 9.90 \times 10^{-4}$	$\mathbf{0.17 \pm 0.05}$	<u>4.19</u>	107.95
	PAN (2024)	0.822 ± 0.019	1.475 ± 0.188	$4.32 \times 10^{-4} \pm 1.29 \times 10^{-4}$	0.34 ± 0.04	5.70	<u>3.57</u>
	Ours	$\mathbf{0.845 \pm 0.016}$	$\underline{1.365 \pm 0.216}$	$1.77 \times 10^{-4} \pm 4.63 \times 10^{-5}$	0.31 ± 0.04	**4.08**	**2.05**
IXI	CorrMLP (2024)	0.806 ± 0.015	2.297 ± 0.316	$2.11 \times 10^{-3} \pm 4.88 \times 10^{-4}$	7.81 ± 0.31	27.77	13.36
	VMambaMorph (2024)	0.828 ± 0.015	$\underline{1.769 \pm 0.271}$	$1.71 \times 10^{-5} \pm 1.35 \times 10^{-5}$	0.57 ± 0.07	19.72	9.64
	MambaMorph (2025)	0.824 ± 0.015	1.840 ± 0.291	$\underline{9.73 \times 10^{-6} \pm 7.22 \times 10^{-6}}$	$\underline{0.52 \pm 0.07}$	11.60	7.59
	TransMorph (2022)	0.830 ± 0.017	2.373 ± 0.321	$1.34 \times 10^{-2} \pm 2.28 \times 10^{-3}$	$\mathbf{0.40 \pm 0.07}$	<u>10.73</u>	107.95
	PAN (2024)	$\underline{0.831 \pm 0.014}$	2.135 ± 0.402	$4.44 \times 10^{-4} \pm 9.16 \times 10^{-5}$	0.80 ± 0.06	14.32	<u>3.57</u>
	Ours	$\mathbf{0.834 \pm 0.015}$	$\mathbf{1.725 \pm 0.281}$	$\mathbf{5.39 \times 10^{-6} \pm 3.54 \times 10^{-6}}$	0.87 ± 0.02	**9.02**	**2.05**

field (Jacobian determinant), computational efficiency (inference time), and resource consumption (peak GPU memory and model size).

As summarised in Table 5, our approach achieves the highest overall accuracy. On OASIS, HCA-Morph attains a Dice score of 0.845 ± 0.016, clearly surpassing CorrMLP (0.831 ± 0.013) and VMambaMorph (0.837 ± 0.015). The advantage persists on IXI, where HCA-Morph achieves 0.834 ± 0.015, again outperforming all competing methods. Boundary accuracy follows the same trend: in terms of HD95, our model records 1.725 ± 0.281 on IXI, markedly lower than CorrMLP (2.297 ± 0.316) and PAN (2.135 ± 0.402), underscoring its superior delineation of fine anatomical structures.

We further evaluated the physical plausibility of the predicted deformation fields using the proportion of voxels with non-positive Jacobian determinants ($|J_\phi \leq 0|$). HCA-Morph maintains exceptionally low values-$1.77 \times 10^{-4} \pm 4.63 \times 10^{-5}$ on OASIS and $5.39 \times 10^{-6} \pm 3.54 \times 10^{-6}$ on IXI-substantially outperforming CorrMLP and TransMorph. These results indicate that the deformation fields produced by HCA-Morph are not only more accurate but also more anatomically plausible, reducing spurious distortions of brain structures.

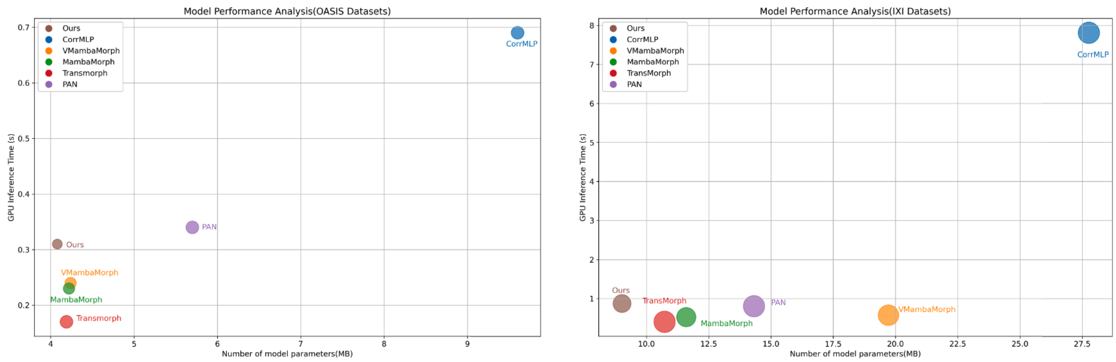
In terms of inference efficiency and model compactness, HCA-Morph achieves a favourable balance between high registration accuracy and practical runtime (see Fig. 8). On the OASIS test set, it completes a full 3D registration in 0.31 ± 0.04 s, which is slightly slower than TransMorph (0.17 ± 0.05 s) yet still well within real-time capability. The model maintains similarly high throughput on IXI, requiring 0.87 ± 0.02 s per image pair. Notably, HCA-Morph is substantially lighter than competing approaches, containing only 2.05 MB of learnable parameters compared with 107.95 MB for TransMorph. This 50× reduction in model size translates into markedly lower memory and storage requirements, greatly facilitating deployment in resource-constrained computing or embedded environments.

Based on the above analysis, although our method maintains leading performance in core metrics such as Dice, the overall margin remains limited. To quantitatively evaluate the reliability of these differences, we report the 95% confidence intervals (CIs) of key metrics on the OASIS and IXI datasets (Table 6). On OASIS, our method exhibits non-overlapping CIs with MambaMorph, TransMorph, CorrMLP, and PAN, indicating statistically significant improvements, whereas the overlap with VMambaMorph suggests no significant difference, although our higher mean and narrower interval imply greater robustness. For HD95, our method performs comparably to TransMorph but surpasses the other baselines, while for the Jacobian determinant it ranks at an intermediate level-better than Transformer and PAN, though not as strong as the Mamba series. On IXI, the Dice results demonstrate statistically significant improvements over TransMorph, CorrMLP, and VMambaMorph, with overlaps observed only against PAN and MambaMorph. For HD95, our method exhibits clear and statistically significant superiority, and for the Jacobian determinant it achieves absolute and statistically significant advantages, being 1-3 orders of magnitude lower than other methods and effectively eliminating non-physical deformations. Overall, these results underscore both the competitive accuracy and robustness of our method, particularly its reliability in preserving anatomically plausible deformations.

To qualitatively evaluate the registration fidelity of individual brain regions, we plotted box-and-whisker diagrams of Dice scores for every anatomical label produced by each method. Fig. 9 reports the results for the 35 labelled structures in the OASIS dataset, whereas Fig. 10 presents the 30 structures in IXI.

Across both datasets, HCA-Morph consistently attains the highest median Dice scores in the majority of regions, demonstrating superior overall accuracy. VMambaMorph and MambaMorph deliver stable performance but lag behind our method in several areas, while TransMorph and PAN exhibit greater score dispersion and more pronounced outliers in certain regions. A closer inspection of challenging structures-such as the ventricles, cerebellum, and cortical margins-reveals that HCA-Morph not only achieves the best Dice accuracy but also yields the smallest interquartile range, reflecting greater robustness. These findings corroborate the effectiveness of the proposed hierarchical attention design (CSAM and SCAM) in modeling multi-level anatomical cues while preserving structural consistency.

On the OASIS dataset (Fig. 9), we further magnified the first seven labels-Left Cerebral White Matter, Left Cerebral Cortex, Left Lateral Ventricle, Left Inferior Lateral Ventricle, Left Cerebellar White Matter, Left Cerebellar Cortex, and Left Thalamus-to examine region-specific performance. For cortical and subcortical tissues (Left Cerebral White Matter/Cortex), HCA-Morph achieves modestly higher median Dice



(a) (b)

Fig. 8. Model complexity and efficiency across the two benchmark datasets. The abscissa shows the number of learnable parameters (MB), the ordinate shows per-pair inference time (s), and the bubble area is proportional to the peak GPU memory footprint during prediction (GB). (a) OASIS test set. (b) IXI test set.

Table 6
95% confidence intervals (CIs) for key metrics on OASIS and IXI. Best results are in **bold**, second best are underlined.

Dataset	Method (Year)	Dice 95% CI ↑	HD95 95% CI ↓	$\lvert J_\phi \le 0\rvert$ 95% CI ↓
OASIS	CorrMLP (2024)	[0.825, 0.837]	[1.421, 1.611]	[8.427×10^{-4}, 1.480×10^{-3}]
	VMambaMorph (2024)	[0.832, 0.854]	[1.371, 1.552]	[1.594×10^{-5}, 1.137×10^{-4}]
	MambaMorph (2025)	[0.817, 0.832]	[1.388, 1.571]	**[6.400×10^{-5}, 8.643×10^{-5}]**
	TransMorph (2022)	[0.810, 0.822]	**[1.242, 1.440]**	[4.383×10^{-3}, 4.813×10^{-3}]
	PAN (2024)	[0.810, 0.835]	[1.360, 1.498]	[3.769×10^{-4}, 4.913×10^{-4}]
	Ours	**[0.837, 0.852]**	[1.275, 1.477]	[1.577×10^{-4}, 1.999×10^{-4}]
IXI	CorrMLP (2024)	[0.806, 0.809]	[2.239, 2.357]	[2.020×10^{-3}, 2.200×10^{-3}]
	VMambaMorph (2024)	[0.819, 0.835]	[1.722, 1.819]	[1.511×10^{-5}, 1.981×10^{-5}]
	MambaMorph (2025)	[0.828, 0.834]	[1.788, 1.894]	[8.451×10^{-6}, 1.107×10^{-5}]
	TransMorph (2022)	[0.827, 0.833]	[2.221, 2.539]	[1.241×10^{-2}, 1.353×10^{-2}]
	PAN (2024)	[0.828, 0.835]	[2.054, 2.149]	[4.198×10^{-4}, 4.972×10^{-4}]
	Ours	**[0.832, 0.836]**	**[1.687, 1.759]**	**[4.951×10^{-6}, 5.858×10^{-6}]**

scores than VMambaMorph and CorrMLP, indicating an improved ability to capture complex boundaries and extensive cortical folding. In the small, deformation-prone ventricular regions, competing methods exhibit wide score dispersion and numerous outliers, whereas HCA-Morph demonstrates a markedly tighter distribution and greater stability. Within the morphologically intricate, low-contrast cerebellum, our model clearly surpasses CorrMLP and TransMorph, underscoring its superior cross-scale anatomical modeling and hierarchical semantic fusion. Finally, for the centrally located thalamus-a structure requiring high spatial fidelity-HCA-Morph attains the highest median Dice score with minimal variance, demonstrating that its correspondence mechanism (SCAM) effectively mitigates misregistration of axial core anatomy.

On the IXI test set (see Fig. 10), we magnified the first seven labels-Left Cerebral Exterior, Left Cerebral White Matter, Left Cerebral Cortex, Left Lateral Ventricle, Left Inferior Lateral Ventricle, Left Cerebellum Exterior, and Left Cerebellum White Matter. These regions span the outer cortical surface, deep white matter, ventricular cavities, and cerebellar margins, all known for ambiguous boundaries, small volumes, and complex morphology. For the Left Cerebral Exterior, which requires precise edge awareness and spatial consistency, HCA-Morph exhibits the most stable performance, markedly reducing low-score outliers compared with CorrMLP and TransMorph. In the core cortical structures (Left Cerebral White Matter/Cortex), our model yields higher median Dice scores than PAN and MambaMorph while presenting fewer outliers, demonstrating superior detail preservation and global alignment. Ventricular regions (Left Lateral Ventricle and Left Inferior Lateral Ventricle) are prone to misregistration owing to their hollow shapes and thin borders; most baselines show substantial variance-CorrMLP even exhibits a pronounced low tail-whereas HCA-Morph maintains a tight, centrally clustered distribution, confirming its robustness in modeling cavity-like anatomy. Finally, in the low-contrast, distal cerebellum (Left Cerebellum Exterior/White Matter), HCA-Morph surpasses CorrMLP and VMambaMorph and achieves the narrowest interquartile ranges, indicating enhanced structural consistency for long-range registration and boundary alignment.

To quantitatively assess registration performance across anatomical structures, we evaluated seven representative regions from the OASIS and IXI datasets and reported three region-wise metrics: sensitivity, specificity, and the worst-case proportion (the fraction of cases with Dice < 0.70). The threshold of 0.70 was selected following Tukey's outlier criterion, where the lower bound $Q_1 - 1.5 \times \mathrm{IQR}$ computed over all Dice scores was approximately 0.68; a slightly more conservative, rounded cutoff of 0.70 was adopted to flag clear statistical outliers. Detailed results are provided in Tables 7 and 8. In terms of sensitivity, the proposed method remains consistently competitive across regions. It achieves the best performance on OASIS for the Left Cerebellar White Matter (0.924) and Left Cerebellar Cortex (0.940), and leads on IXI for the Left Cerebral Exterior (0.889) and Left Cerebral White Matter (0.794), indicating strong alignment of both cortical and subcortical structures. Specificity remains near perfect for all methods (typically ≈ 1.000), confirming accurate spatial localization with minimal false positives. Most notably, the proposed method demonstrates superior robustness: it achieves the lowest or tied-lowest worst-case proportions in nearly all regions, markedly reducing failure rates in challenging structures such as the Left Inferior Lateral Ventricle on OASIS (0.238 vs.

Table 7
Regional sensitivity, specificity, and worst-case proportion (Dice < 0.70) across different regions in OASIS. Arrows indicate the desired direction of each metric (↑ higher is better, ↓ lower is better).

Method	Metric	Left Cerebral White Matter	Left Cerebral Cortex	Left Lateral Ventricle	Left Inferior Lateral Ventricle	Left Cerebellar White Matter	Left Cerebellar Cortex	Left Thalamus
CorrMLP	Sensitivity ↑	0.724	0.548	0.915	0.689	0.895	0.883	0.935
	Specificity ↑	0.989	0.980	**1.000**	**1.000**	**1.000**	**0.999**	**1.000**
	Worst ↓	0.526	1.000	**0.000**	0.474	**0.000**	**0.000**	**0.000**
VMambaMorph	Sensitivity ↑	0.894	0.793	**0.945**	**0.740**	0.904	0.915	0.935
	Specificity ↑	0.995	0.990	**1.000**	**1.000**	**1.000**	**0.999**	**1.000**
	Worst ↓	**0.000**	**0.000**	**0.000**	0.368	**0.000**	**0.000**	**0.000**
MambaMorph	Sensitivity ↑	0.888	0.786	0.943	0.734	0.887	0.913	0.939
	Specificity ↑	0.995	0.990	**1.000**	**1.000**	**1.000**	**0.999**	**1.000**
	Worst ↓	**0.000**	**0.000**	**0.000**	0.579	**0.000**	**0.000**	**0.000**
TransMorph	Sensitivity ↑	0.885	0.716	0.933	0.550	0.892	0.910	0.930
	Specificity ↑	0.993	0.988	**1.000**	**1.000**	**1.000**	**0.999**	**1.000**
	Worst ↓	**0.000**	0.158	**0.000**	0.842	**0.000**	**0.000**	**0.000**
PAN	Sensitivity ↑	**0.925**	**0.827**	0.936	0.730	0.902	0.918	0.929
	Specificity ↑	**0.996**	**0.992**	**1.000**	**1.000**	**1.000**	**0.999**	**1.000**
	Worst ↓	**0.000**	**0.000**	**0.000**	0.368	**0.000**	**0.000**	**0.000**
Ours	Sensitivity ↑	0.897	0.803	0.944	0.681	**0.924**	**0.940**	**0.954**
	Specificity ↑	**0.996**	0.990	**1.000**	**1.000**	**1.000**	**0.999**	**1.000**
	Worst ↓	**0.000**	**0.000**	**0.000**	**0.238**	**0.000**	**0.000**	**0.000**

Table 8
Regional sensitivity, specificity, and worst-case proportion (Dice < 0.70) across different regions in IXI. Arrows indicate the desired direction of each metric (↑ higher is better, ↓ lower is better).

Method	Metric	Left Cerebral Exterior	Left Cerebral White Matter	Left Cerebral Cortex	Left Lateral Ventricle	Left Inf Lat Vent	Left Cerebellum Exterior	Left Cerebellum White Matter
CorrMLP	Sensitivity ↑	0.753	0.541	0.902	0.854	0.846	**0.922**	**0.896**
	Specificity ↑	0.981	0.975	**1.000**	**1.000**	0.998	**1.000**	**1.000**
	worst ↓	0.757	1.000	**0.000**	**0.000**	0.017	**0.000**	0.009
VMambaMorph	Sensitivity ↑	0.854	0.786	0.894	0.838	0.908	0.920	0.883
	Specificity ↑	**0.993**	0.982	**1.000**	**1.000**	**0.999**	**1.000**	**1.000**
	worst ↓	**0.000**	0.139	**0.000**	0.017	**0.000**	**0.000**	0.009
MambaMorph	Sensitivity ↑	0.833	0.655	0.838	0.805	0.883	0.892	0.824
	Specificity ↑	0.988	0.980	**1.000**	**1.000**	0.998	**1.000**	**1.000**
	worst ↓	**0.000**	1.000	0.078	0.009	0.009	**0.000**	0.070
TransMorph	Sensitivity ↑	0.868	0.760	0.915	0.854	0.888	0.915	0.873
	Specificity ↑	0.992	**0.983**	**1.000**	**1.000**	0.998	**1.000**	**1.000**
	worst ↓	**0.000**	0.157	**0.000**	**0.000**	0.009	**0.000**	0.035
PAN	Sensitivity ↑	0.860	0.789	0.917	**0.864**	**0.911**	0.912	0.875
	Specificity ↑	0.992	**0.983**	**1.000**	**1.000**	0.998	**1.000**	**1.000**
	worst ↓	**0.000**	0.104	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**
Ours	Sensitivity ↑	**0.889**	**0.794**	**0.920**	0.859	0.903	0.905	0.880
	Specificity ↑	**0.993**	0.958	**1.000**	**1.000**	**0.999**	**1.000**	**1.000**
	worst ↓	**0.000**	**0.026**	**0.000**	**0.000**	**0.000**	**0.000**	0.009

0.368–0.842 for other methods) and the Left Cerebral White Matter on IXI (0.026 vs. 0.104–1.000). These findings demonstrate a substantial reduction in severe misregistrations.

Fig. 11 presents qualitative registration results on both the IXI and OASIS datasets, showing the fixed image, moving image, predicted flow field, and resulting warped image for each method. The flow-field visualizations reveal that HCA-Morph produces the largest deformations and the most pronounced grid distortions, indicating a superior ability to establish long-range voxel correspondences-an essential property when large structural displacements or organ shifts are involved. Consistently, the warped images generated by HCA-Morph exhibit contours, boundary details, and anatomical overlaps that most closely align with the fixed image, reflecting higher anatomical fidelity and registration accuracy. Although VMambaMorph and MambaMorph capture several structures reasonably well, their overall deformation magnitude is limited, leaving noticeable misalignments in multiple regions. TransMorph and PAN display relatively weaker performance in boundary-critical areas such as the Left Globus Pallidus and Right Putamen, where substantial mismatches remain. These qualitative observations reinforce the quantitative findings: HCA-Morph not only achieves the highest overall accuracy but also demonstrates superior deformation expressiveness and structural alignment, highlighting its effectiveness for challenging brain MRI registration tasks.

The experimental results demonstrate that HCA-Morph achieves systematic and statistically significant improvements across all evaluation dimensions. While the absolute gains in Dice scores appear numerically modest (+0.008 on OASIS, +0.003 on IXI), the non-overlapping 95% confidence intervals (Table 6) confirm these differences are statistically meaningful rather than measurement noise. More importantly, these improvements are consistent across all 30 anatomical structures (Tables 7-8), both datasets, and multiple complementary metrics-indicating genuine algorithmic advantages rather than dataset-specific overfitting. Beyond conventional overlap metrics, HCA-Morph achieves substantially superior deformation quality: the fraction of topologically implausible transformations ($|J_\phi \leq 0|$) is reduced by 1–3 orders of magnitude compared to all baselines–from $[1.241 \times 10^{-2}, 1.353 \times 10^{-2}]$ (Trans-

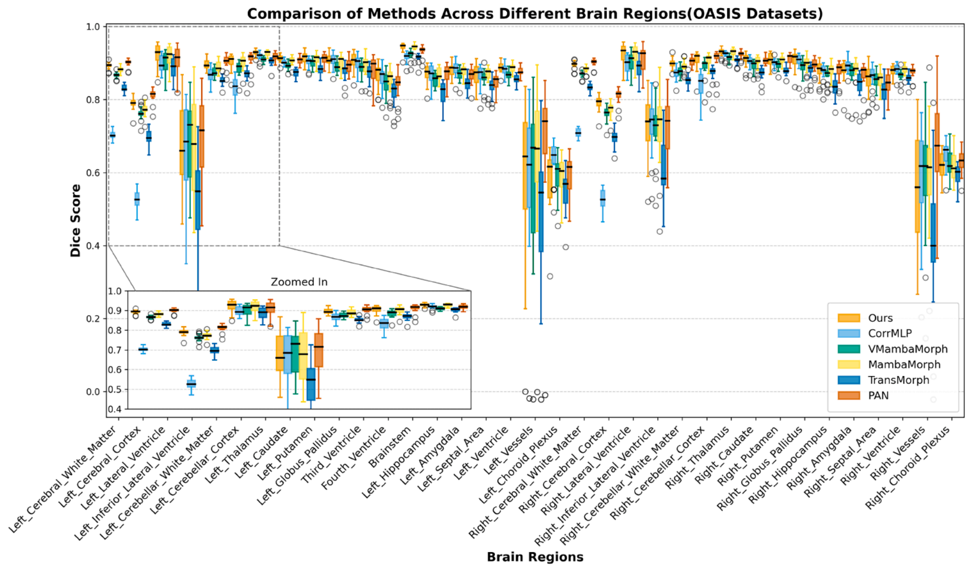


Fig. 9. Boxplots of Dice scores for each anatomical region on the OASIS dataset, comparing the proposed method with all baseline models.

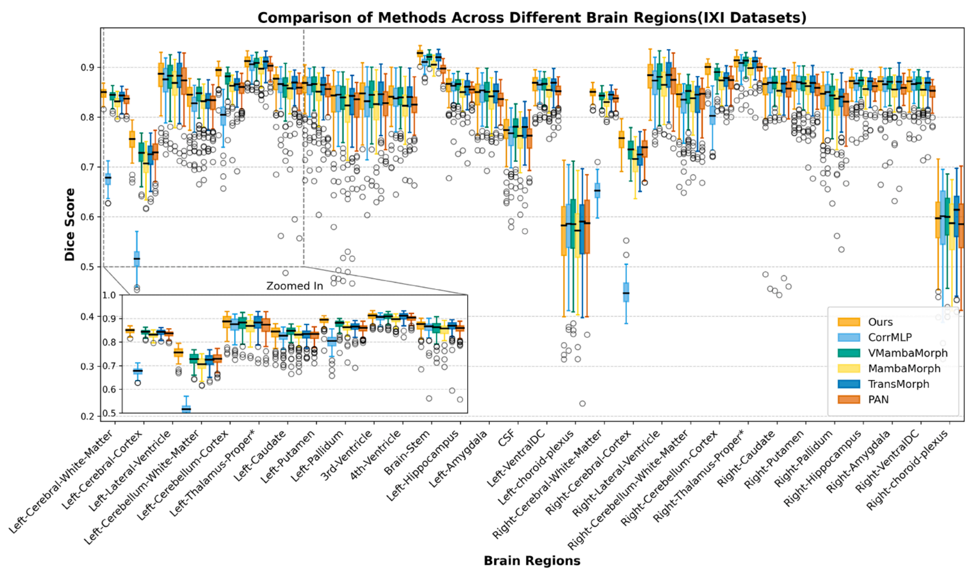


Fig. 10. Boxplots of Dice scores for each anatomical region on the IXI dataset, comparing the proposed method with all baseline models.

Morph) to $[4.951 \times 10^{-6}, 5.858 \times 10^{-6}]$ (ours) on IXI. This translates directly to more reliable registrations in anatomically challenging regions: worst-case failure rates (Dice < 0.70) drop from 0.842 to 0.238 in the left inferior lateral ventricle on OASIS, while maintaining near-perfect boundary precision (HD95 reductions of 5–8% across datasets). Critically, these quality improvements are achieved with the lowest computational footprint among all methods–4.08 GB memory on OASIS (42% reduction vs. VMambaMorph) and only 2.05 MB parameters-making HCA-Morph uniquely suited for deployment in resource-constrained clinical settings where both accuracy and efficiency are essential.

3.5.2. Ablation study

To quantify the individual contributions of the Spatial Correspondence-Aware Module (SCAM) and the Cross-Scale Attention Module (CSAM), as well as to examine their potential synergistic effects, we conducted ablation experiments on both the OASIS and IXI datasets. Four configurations were evaluated: (i) SCAM only, (ii) CSAM only, (iii) neither module, and (iv) the full model (HCA-Morph, SCAM + CSAM).

To quantitatively elucidate the individual contributions of the SCAM and CSAM modules and to evaluate their potential synergistic effects, Table 9 shows that across both datasets, the SCAM-only model consistently yields a lower proportion of voxels with non-positive Jacobian determinants compared with the CSAM-only model (OASIS: 1.41×10^{-2} vs. 1.55×10^{-2}; IXI: 1.35×10^{-2} vs. 1.62×10^{-2}). This indicates that SCAM provides clearer voxel-wise correspondences, tighter alignment, and stronger local geometric regularization. Meanwhile, the CSAM-only model achieves superior HD95 values compared with the SCAM-only model (OASIS: 1.403 vs. 1.464; IXI: 2.001 vs. 6.483), suggesting that leveraging high-level contextual information to guide low-level representations enhances boundary correspondence accuracy and improves global cross-scale consistency. The complete HCA-Morph (SCAM + CSAM) combines these strengths and compensates for their respective limitations, achieving the best Dice and HD95 scores on both datasets (OASIS: 0.845, 1.365; IXI: 0.834, 1.725) while also attaining the lowest proportion of non-positive Jacobian determinants (OASIS: 1.77×10^{-4}; IXI: 5.39×10^{-6}), thereby demonstrating the synergistic interplay between SCAM and CSAM.

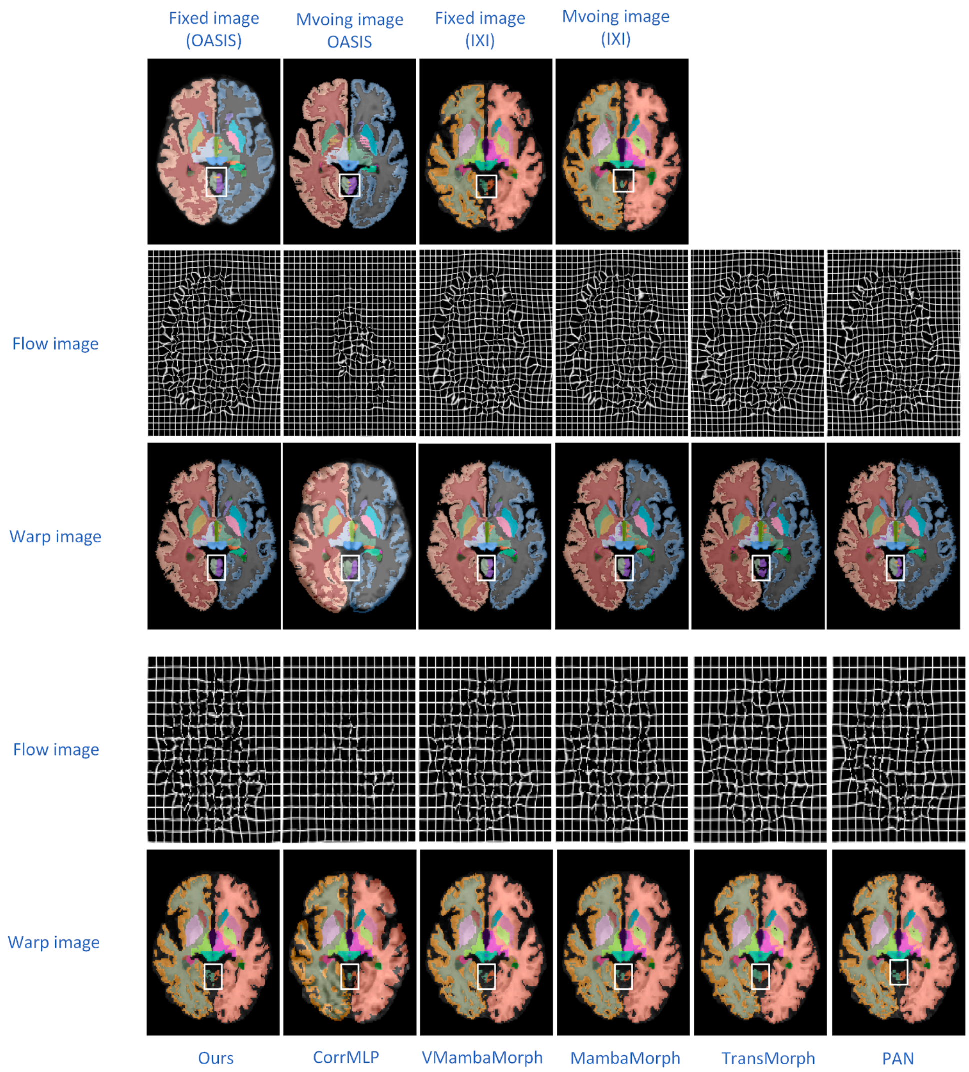


Fig. 11. Qualitative results on the IXI and OASIS datasets. Flow visualises the deformation applied to the moving image; Warp is the moving image after registration. Colour-coded anatomical structures highlight alignment quality across methods. For clarity, the white boxes indicate the Left Globus Pallidus and Right Putamen.

Table 9
Ablation study of HCA-Morph components on the OASIS and IXI test datasets. Best results are in **bold**.

Dataset	Method	Dice ↑	HD95 ↓	$\lvert J_\phi \le 0\rvert$ ↓	Time (s) ↓
OASIS	SCAM-only	0.819 ± 0.012	1.464 ± 0.195	$1.41 \times 10^{-2} \pm 1.87 \times 10^{-3}$	0.30 ± 0.06
	CSAM-only	0.821 ± 0.013	1.403 ± 0.199	$1.55 \times 10^{-2} \pm 2.29 \times 10^{-3}$	0.29 ± 0.07
	NONE	0.815 ± 0.014	2.654 ± 0.491	$1.16 \times 10^{-2} \pm 2.23 \times 10^{-3}$	$\mathbf{0.29 \pm 0.05}$
	HCA	$\mathbf{0.845 \pm 0.016}$	$\mathbf{1.365 \pm 0.216}$	$\mathbf{1.77 \times 10^{-4} \pm 4.63 \times 10^{-5}}$	0.31 ± 0.04
IXI	SCAM-only	0.830 ± 0.016	6.483 ± 0.672	$1.35 \times 10^{-2} \pm 3.29 \times 10^{-3}$	0.74 ± 0.06
	CSAM-only	0.833 ± 0.016	2.001 ± 0.363	$1.62 \times 10^{-2} \pm 3.31 \times 10^{-3}$	0.74 ± 0.07
	NONE	0.824 ± 0.015	1.840 ± 0.291	$9.73 \times 10^{-6} \pm 7.22 \times 10^{-6}$	$\mathbf{0.52 \pm 0.07}$
	HCA	$\mathbf{0.834 \pm 0.015}$	$\mathbf{1.725 \pm 0.281}$	$\mathbf{5.39 \times 10^{-6} \pm 3.54 \times 10^{-6}}$	0.87 ± 0.02

Compared with the NONE ablation model, the inclusion of either SCAM or CSAM results in higher Jacobian determinant values, which may arise from the use of explicit channel concatenation for spatial correspondence fusion, the pyramid-based paradigm for multi-level feature integration, or the combined effects of SCAM-CSAM interactions. In subsequent qualitative analyses, we further observe that employing CSAM alone weakens inter-level feature attention. Moreover, as shown in the Module-Integration Experiments (Section 3.5.3), integrating SCAM or CSAM with simpler or less context-aware backbones can also degrade overall performance. Regarding inference time, the differences across ablation configurations are marginal, indicating that the proposed modules do not introduce significant computational overhead and remain compatible with time-sensitive scenarios.

To further elucidate the respective roles of SCAM and CSAM and to examine whether a genuine synergistic effect exists, interpretability analyses were conducted for each module and validated on both datasets. For SCAM, whose primary function is to localize in the fixed image the anatomical sites corresponding to structures in the moving image and thereby facilitate spatial alignment, two anatomically salient regions were analyzed: the Left Cerebellar Cortex on OASIS

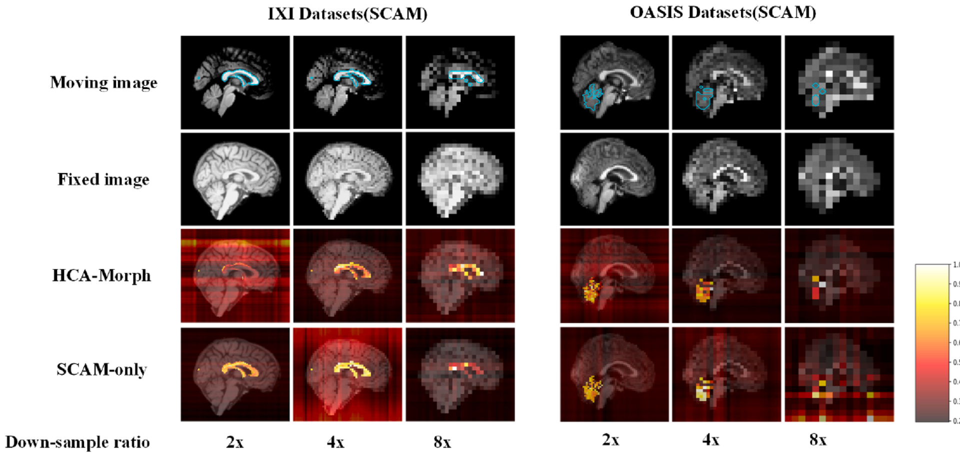


Fig. 12. The heatmap of the interpretability analysis results for the SCAM module on the IXI and OASIS datasets. Color brightness corresponds to the magnitude of the contribution. The Moving Image represents feature maps with segmentation masks added at different levels; the Fixed Image represents the feature maps of the Fixed image at different levels. The figure compares the HCA-Morph and SCAM-only methods, where each row displays the attention heatmaps of fixed feature maps at different levels after being processed by the SCAM module. The notations 2x, 4x, and 8x denote the multi-level feature maps obtained with different downsampling rates, i.e., feature maps from different hierarchical levels.

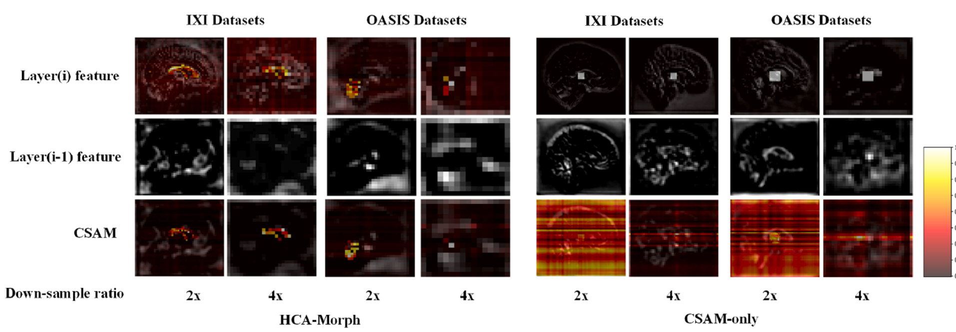


Fig. 13. The heatmap of the interpretability analysis results for the CSAM module on the IXI and OASIS datasets. Color brightness corresponds to the magnitude of the contribution. The Layer(i) features denote the output feature maps of the moving and fixed images at the ith layer (lower downsampling rate), while the Layer(i-1) features correspond to the output feature maps at the (i-1)th layer (higher downsampling rate). CSAM represents the result of applying attention from the Layer(i) features to the Layer(i-1) features (i.e., the CSAM module). The notations 2x, 4x, and 8x indicate feature maps at different hierarchical levels obtained with varying downsampling rates.

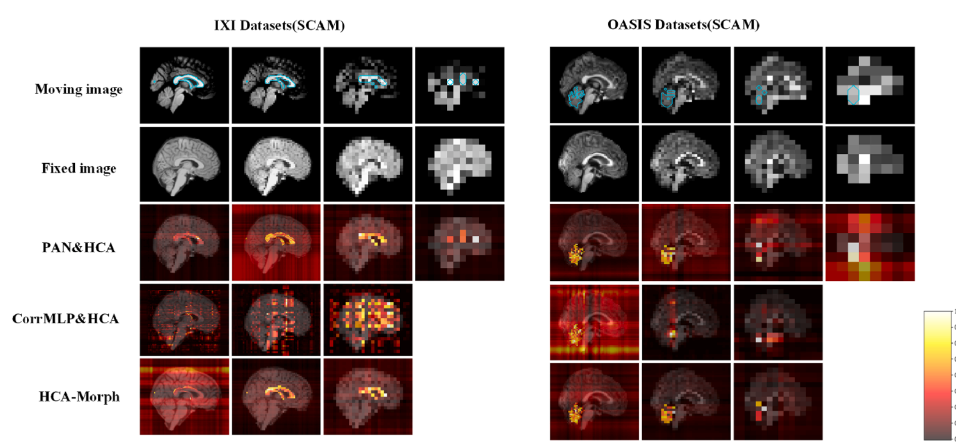


Fig. 14. Interpretability heatmaps of the SCAM module when integrated into PAN and CorrMLP on the IXI and OASIS datasets. Color brightness indicates the magnitude of contribution. The moving image rows show feature maps with segmentation masks added at different levels, while the fixed image rows depict the corresponding fixed-image features. Each subsequent row presents attention heatmaps of fixed features after processing with SCAM at varying downsampling ratios (2×, 4×, 8×), representing multi-level feature maps extracted from different hierarchical levels.

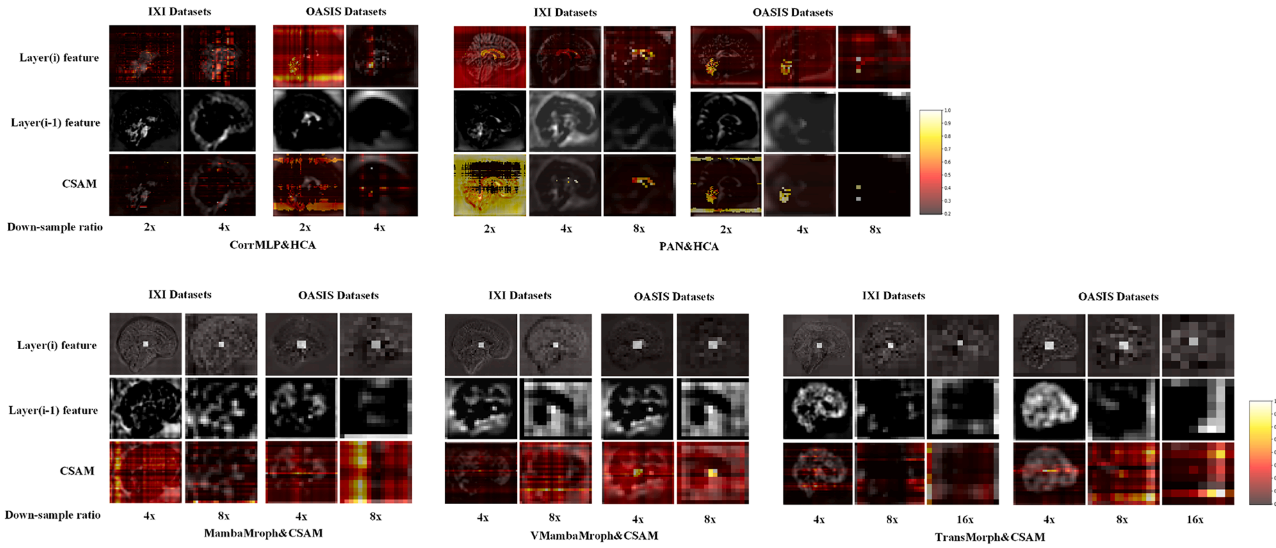


Fig. 15. Interpretability heatmaps of the CSAM module when integrated into CorrMLP, PAN, MambaMorph, VMambaMorph, and TransMorph on the IXI and OASIS datasets. Color brightness indicates the magnitude of contribution. The Layer(i) features denote the output feature maps of moving and fixed images at the ith layer (lower downsampling rate), while the Layer($i-1$) features correspond to the output of the $(i-1)$th layer (higher downsampling rate). CSAM represents the result of applying attention from Layer(i) features to Layer($i-1$) features, i.e., the CSAM operation. The notations 2×, 4×, 8×, and 16× indicate feature maps obtained at different hierarchical levels with varying downsampling rates.

Table 10
Performance impact of integrating CSAM and HCA into baseline models on the OASIS and IXI test sets.

Dataset	Method	Module	Dice ↑	HD95 ↓	$\lvert J_\phi \leq 0\rvert$ ↓
OASIS	CorrMLP	HCA	0.814±0.013	1.550±0.220	$2.30 \times 10^{-3} \pm 1.23 \times 10^{-3}$
	VMambaMorph	CSAM	0.842±0.013	1.379±0.013	$9.19 \times 10^{-6} \pm 4.08 \times 10^{-6}$
	MambaMorph	CSAM	0.833±0.012	1.417±0.224	$5.87 \times 10^{-6} \pm 3.22 \times 10^{-6}$
	TransMorph	CSAM	0.811±0.012	1.407±0.215	$5.14 \times 10^{-3} \pm 9.72 \times 10^{-4}$
	PAN	HCA	0.827±0.014	1.371±0.213	$5.53 \times 10^{-3} \pm 1.21 \times 10^{-3}$
IXI	CorrMLP	HCA	0.706±0.025	3.581±0.434	$1.51 \times 10^{-3} \pm 2.61 \times 10^{-4}$
	VMambaMorph	CSAM	0.833±0.016	1.905±0.324	$3.01 \times 10^{-5} \pm 1.93 \times 10^{-5}$
	MambaMorph	CSAM	0.833±0.015	1.833±0.286	$7.38 \times 10^{-6} \pm 5.01 \times 10^{-6}$
	TransMorph	CSAM	0.820±0.016	2.009±0.312	$8.50 \times 10^{-3} \pm 1.95 \times 10^{-3}$
	PAN	HCA	0.828±0.015	1.968±0.402	$1.42 \times 10^{-2} \pm 3.89 \times 10^{-3}$

and the Left Cerebral White Matter on IXI. Attention responses were tracked across successive downsampling stages to determine whether the moving-region of interest was consistently highlighted within the fixed-image features. As shown in Fig. 12, image detail degrades with increasing downsampling (e.g., at 8× on OASIS, the moving region becomes nearly indistinguishable), yet SCAM maintains high-confidence localization in the fixed-image attention maps. On IXI, HCA-Morph exhibits slightly weaker tracking at 2× but still delineates the structure, with attention concentrated in the superior portion of the fixed image. Importantly, Table 9 shows that such local variability does not compromise the overall performance of HCA-Morph, supporting the presence of SCAM-CSAM synergy. By contrast, the SCAM-only model displays degraded tracking at 8× on OASIS, plausibly reflecting the absence of cross-scale semantic guidance required to model feature distributions at high downsampling. In summary, SCAM emphasizes organ-level correspondences while exhibiting limited global contextualization, and effectively drives spatial alignment between moving and fixed images.

For CSAM, high-level (low downsampling rate) features leverage the global context provided by low-level (high downsampling rate) features, thereby enabling effective multi-level fusion. In HCA-Morph, the presence of SCAM provides reliable alignment cues, making it possible to directly visualize organ-centered attention and verify whether the corresponding regions are accurately retrieved at low-level scales. In the CSAM-only setting, since the moving and fixed images are fused through simple channel concatenation for spatial correspondence, no effective reference points are available during multi-level feature fusion. To determine whether CSAM functions as intended, we introduced a central binary mask into the fused feature maps after spatial correspondence fusion to reveal cross-scale behavior, observing how seed points at high-level scales map to key locations at low-level scales (Fig. 13). HCA-Morph consistently identifies the correct regions across both datasets, whereas CSAM-only produces less satisfactory results: although it occasionally localizes key positions at high-level scales, it also highlights substantial amounts of irrelevant information. These observations align with Table 9: compared with CSAM-only (which provides stronger local geometric regularization), SCAM-only achieves a lower proportion of voxels with non-positive Jacobians, while CSAM-only attains lower HD95 values, thereby reducing worst-case boundary errors. Only their combination can simultaneously anchor voxel-wise correspondences and propagate global context, resulting in superior overall registration performance.

In summary, the ablation and interpretability results reveal complementary yet incomplete behaviors for the single-module configurations, and a consistent advantage when both modules are combined. SCAM-only provides sharper voxel-wise anchoring and smaller fractions of voxels with $\lvert J_\phi \leq 0\rvert$ than CSAM-only, yet can exhibit degraded boundary accuracy on IXI (larger HD95). CSAM-only achieves lower HD95 and modest Dice improvements relative to SCAM-only but incurs higher folding rates and less precise localization. The combined model (HCA-Morph) attains the best Dice and HD95 scores together with the lowest $|J_\phi \leq 0|$ on both OASIS and IXI, while introducing only minimal runtime overhead (approximately +0.02 s on OASIS and +0.35 s on IXI). Crucially, the interpretability analyses (Figs. 12-13) demonstrate that the integration of SCAM and CSAM enables mechanistic transparency-a capability absent in existing black-box registration networks. SCAM maintains robust anatomical localization even under severe (8×) downsampling by establishing explicit spatial correspondences, while CSAM-only highlights broader irrelevant regions without such anchoring. When combined, SCAM first anchors anatomically salient structures through organ-level correspondence modeling, and CSAM then propagates global context across scales using these reliable reference points. This interpretable two-stage mechanism-spatial anchoring followed by context-aware fusion-not only explains the quantitative superiority in Table 9 but also allows users to trace how specific anatomical features influence the final deformation field, addressing a fundamental limitation of end-to-end deep learning approaches where internal decision-making processes remain opaque.

3.5.3. Module-integration experiments

To evaluate the portability and performance gains introduced by the CSAM and HCA modules, we integrated them into five mainstream registration backbones-CorrMLP, VMambaMorph, MambaMorph, TransMorph, and PAN-and retrained each model on the OASIS and IXI datasets.

Most backbone networks substantially benefited from the integrated modules (see Table 10), with the largest improvements observed in the two Mamba-based variants. On the OASIS dataset, VMambaMorph with CSAM increased Dice from 0.837 to 0.842 (↑ 0.005) and reduced HD95 from 1.433 to 1.379 (↓ 0.054). Similarly, MambaMorph with CSAM improved Dice from 0.826 to 0.833 (↑ 0.007) and decreased HD95 from 1.521 to 1.417 (↓ 0.104). On the IXI dataset, both models achieved additional Dice improvements of 0.005–0.009, accompanied by consistent reductions in HD95, confirming that CSAM can be seamlessly integrated into Mamba-style encoders. PAN with HCA also benefited from the full HCA module, achieving higher Dice scores and lower HD95 values.

In contrast, the integration results for TransMorph and CorrMLP were less favorable. Although HD95 improved modestly, Dice performance declined. For TransMorph, Dice dropped by 0.005 on OASIS and 0.010 on IXI, despite a notable HD95 reduction (e.g., from 2.373 to 2.009 on IXI). CorrMLP exhibited the most pronounced degradation, with Dice decreases of 0.017 on OASIS and 0.100 on IXI, accompanied by substantial increases in HD95.

To further investigate why HCA and CSAM exhibited divergent integration effects across models, we visualized the attention maps of the SCAM and CSAM modules within different architectures. Fig. 14 illustrates the SCAM-related results for PAN and CorrMLP, with HCA-Morph included for comparison. Both PAN and HCA-Morph effectively captured correspondences between the moving and fixed images in regions such as the Left Cerebellar Cortex and Left Cerebral White Matter on OASIS and IXI, whereas CorrMLP exhibited less reliable attention. This limitation likely stems from the relatively simple feature extraction design of CorrMLP, which relies solely on convolution and downsampling.

Fig. 15 presents the CSAM integration results for PAN, CorrMLP, MambaMorph, VMambaMorph, and TransMorph. CorrMLP, which already underperformed at the SCAM stage, struggled to establish consistent correspondences between the moving and fixed images, leading to weak cross-scale interactions. TransMorph also showed limited improvement, likely due to disrupted spatial context caused by direct channel-wise fusion and overly aggressive downsampling. MambaMorph achieved only coarse localization of salient regions in highly downsampled feature maps, although Table 10 indicates that it still achieved notable quantitative gains. In contrast, the remaining models successfully leveraged CSAM to highlight salient regions in higher-resolution feature maps, yielding consistent improvements in Dice and related metrics.

In summary, the module-integration experiments confirm that CSAM and HCA possess strong portability across heterogeneous backbones and provide consistent incremental benefits in registration accuracy. The results highlight that the SCAM component effectively facilitates correspondence learning between the moving and fixed images, while the CSAM component enhances cross-scale feature interactions by dynamically integrating information across different resolution levels. The most substantial improvements are observed in Mamba-based models, which are particularly well-suited to exploit these modules, whereas simpler or less context-aware backbones such as CorrMLP and TransMorph exhibit limited or inconsistent gains. These findings suggest that while the proposed modules are generally applicable, their benefits depend on the representational capacity and architectural design of the host network, with Mamba-based backbones showing the greatest compatibility and performance gains.

4. Conclusion

We introduced HCA-Morph, a deformable registration framework that integrates a Spatial Correspondence-Aware Module (SCAM) and a Cross-Scale Attention Module (CSAM). By coupling explicit voxel-wise correspondence modeling with adaptive multi-level feature fusion, HCA-Morph achieves superior accuracy, robustness, and efficiency compared with state-of-the-art baselines. Experiments on the OASIS and IXI datasets demonstrate consistent improvements in Dice, HD95, and Jacobian determinant rate, while maintaining a lightweight model design. Critically, SCAM and CSAM possess inherent mechanistic interpretability: SCAM's explicit spatial correspondence modeling enables direct visualization of anatomical anchoring across hierarchical levels, while CSAM's cross-scale fusion mechanism transparently reveals how global context propagates to guide feature refinement. This built-in interpretability-allows users to trace how specific anatomical structures influence the final deformation field, addressing a fundamental limitation of end-to-end deep learning approaches. Ablation and integration studies further confirm that SCAM and CSAM offer complementary benefits and can be seamlessly ported across diverse architectures, demonstrating both their effectiveness and generalizability.

Nonetheless, several limitations remain. Performance may degrade in anatomically ambiguous regions, occasional anatomically implausible deformations can still occur, and portability is partly constrained by the representational capacity of the host backbone. Moreover, validation has thus far been limited to T1-weighted brain MRI. Future work may include extending HCA-Morph to multi-modality data, enhancing interpretability, and applying it to other anatomical domains. In addition, although our model substantially reduces parameter count and memory footprint, further optimization for deployment on low-power or edge devices-often the practical computational resources in clinical settings-warrants exploration. Beyond medical image registration, the proposed strategy of combining explicit correspondence modeling with adaptive cross-scale fusion may also inspire advances in related areas such as natural image alignment, remote sensing, and cross-modal representation learning. Overall, HCA-Morph offers a principled and efficient framework for deformable registration.

CRediT authorship contribution statement

Dongdong Wan: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization; **Jiaxuan Jiang:** Data curation; **Yuee Li:** Methodology; **Wenyan Li:** Resources; **Zhong Wang:** Writing – original draft, Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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